Attention-Enhanced Spatio-Temporal Deep Learning Model for Solar Irradiance Forecasting

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Abstract

Accurate short-term forecasting of Global Horizontal Irradiance (GHI) plays a vital role in ensuring the reliable integration of solar power into electrical grids. This study introduces a hybrid, site-specific forecasting framework that merges physically based clear-sky modelling with a deep learning model capable of representing cloud-induced variability. The methodology utilises visible-band INSAT satellite imagery to construct high-resolution spatiotemporal cloud mask sequences. The central contribution is a hybrid deep learning architecture that combines a Convolutional Long Short-Term Memory (ConvLSTM) Autoencoder with a Multi-Head Attention mechanism, designed to capture and compress predictive patterns from sequential cloud observations. The latent features are then fused with clear-sky GHI, derived from the reduced Solis model, to achieve robust multi-step GHI forecasting. The model is trained using a sliding-window approach, employing six previous observations to predict a two-hour (four-step) horizon. Validation using ground-truth data from Tirupati, India, yielded a Mean Absolute Error (MAE) of 110.98 and a Root Mean Square Error (RMSE) of 142.38. The proposed hybrid model demonstrates high potential for operational nowcasting systems, particularly in scenarios where satellite data are the primary input source.

Index Terms—Attention mechanism, cloud-aware, ConvLSTM, deep learning, GHI prediction, short-term solar forecasting, spatiotemporal autoencoder.

I. Introduction

As the global energy landscape shifts toward renewable alternatives, solar power has emerged as one of the most promising and sustainable resources. However, the inherently intermittent nature of solar radiation poses significant challenges for grid stability and energy management. A critical variable in addressing this challenge is Global Horizontal Irradiance (GHI)—the total solar energy incident on a horizontal surface per unit area. Accurate GHI forecasting is essential for optimising photovoltaic (PV) system operation, maintaining grid reliability, and supporting energy dispatch planning.

Traditional GHI prediction methods predominantly rely on Numerical Weather Prediction (NWP) models. While these models perform adequately for large-scale or long-term forecasting, they often lack the fine spatial and temporal resolution necessary for short-term "nowcasting" (0–6 hours ahead). Consequently, NWP models struggle to capture the rapid and localized variations in cloud cover that primarily drive GHI fluctuations.

In recent years, data-driven approaches such as Artificial Neural Networks (ANNs) have been used for solar forecasting. However, most classical models treat data as static and fail to represent the complex spatiotemporal evolution of cloud dynamics. Modelling how cloud formations evolve and move over time is critical for improving forecast accuracy.

Spatiotemporal deep learning models—especially Convolutional Long Short-Term Memory (ConvLSTM) networks—have shown strong capabilities in learning from sequential image data. ConvLSTMs effectively merge the spatial feature extraction strength of Convolutional Neural Networks (CNNs) with the temporal modelling ability of LSTMs.

Despite these advantages, relatively few frameworks integrate ConvLSTM-based deep learning with the physical principles governing clear-sky irradiance. This study addresses that gap by proposing a novel hybrid model that fuses a data-driven ConvLSTM Autoencoder with a physics-based clear-sky GHI model, enabling more accurate and physically consistent forecasts.

The main contributions of this paper are as follows:

- The design of a hybrid forecasting architecture that integrates a ConvLSTM Autoencoder with the physics-based Solis clear-sky model.
- The use of visible-band INSAT satellite imagery to support GHI forecasting over data-scarce regions in India.

- A comprehensive performance evaluation of the model for two-hour, multi-step solar irradiance forecasting.

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II. Data Sources and Proposed Methodology

This section describes the data sources, preprocessing steps, and the architecture of the proposed hybrid forecasting model.

A. Data Sources and Preprocessing

1) Ground-Truth Data: GHI measurements for model training and validation were obtained from a ground-based pyranometer located at Tirupati, India (13.627° N, 79.397°E).

2) Satellite Data: Spatiotemporal cloud data was sourced from the Indian National Satellite System (INSAT) series, operated by the Indian Space Research Organisation (ISRO). We utilised visible band images with a 30-minute temporal resolution. The raw data, provided in HDF5 format, covers the Indian subcontinent.

3) Physics-Based Data: A theoretically clear-sky GHI baseline was generated using the pvlib Python library, a standard tool in solar energy analysis. We employed the reduced Solis clear-sky model, which calculates irradiance based on the geographic location and solar position (e.g., zenith and azimuth angles).

Data preprocessing involved extracting the visible band frames from the HDF5 files, clipping the images to a 256x256 pixel Area of Interest (AOI) centered on the Tirupati ground station, and normalising pixel intensities to a range of 0 to 1 to ensure consistency during model training

B. Proposed Hybrid Forecasting Framework

The core of our methodology is a novel hybrid frame-work that fuses a spatiotemporal deep learning model with a physical solar model. The system architecture is illustrated in Fig.1.

1) Spatiotemporal Autoencoder Design: The encoder employs multiple Conv2D layers to extract spatial features like cloud edges and texture patterns, followed by pooling for spatial downsampling. The decoder mirrors this structure using Conv2DTranspose layers to reconstruct the images from latent features.

• Encoder: The encoder consists of a series of 2D Convolutional (Conv2D) layers. These layers apply filters to the input image sequence, learning to extract spatial features such as cloud edges, textures, and shapes. Pooling layers are used to progressively downsample the spatial dimensions, creating an increasingly abstract and compressed feature map.

• Decoder: The decoder mirrors the encoder, using Transposed Convolutional (Conv2DTranspose) layers to upsample the feature maps back to their original 256x256 resolution. This process reconstructs the image from the compressed latent representation.

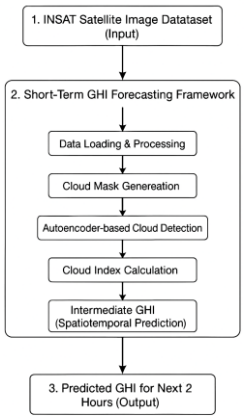

Fig. 1.High-level architecture diagram of the proposed GHI prediction framework, showing the integration of INSAT data, the Autoencoder, the ConvLSTM model, and the final GHI calculation.

2) ConvLSTM Integration Strategy: A ConvLSTM layer is placed at the bottleneck of the autoencoder to capture temporal evolution in the compressed spatial features. This integration allows the model to understand cloud movement and dissipation over time, providing dynamic context rather than static analysis. The ConvLSTM layer processes the sequence of compressed feature maps from the encoder, allowing it to learn the temporal evolution of the spatial features. This enables the model to understand cloud motion, formation, and dissipation over time, rather than just learning static cloud patterns.

3) Model Enhancements: Layer Normalisation and Attention: To improve gradient flow and model stability, Layer Normalization is applied after the ConvLSTM layers. Furthermore, a Multi-Head Attention mechanism (with 4 heads and a key dimension of 16) is integrated after the temporal feature extraction. This allows the model to weigh the importance of different time steps in the input sequence, enhancing its ability to capture complex temporal dependencies, especially during rapid cloud transitions. A Dropout layer with a rate of 0.3 is also added after the dense layer to improve generalization and prevent overfitting.

C. GHI Prediction Pipeline

1) Cloud Mask Generation: The trained autoencoder is first used to generate cloud masks. By analyzing the reconstruction error between an input satellite image and the autoencoder's output, the model can identify pixels corresponding to clouds (which typically have a higher reconstruction error) versus clear sky.

2) Cloud Index (CI) Computation: The generated cloud mask is converted into a numerical Cloud Index(CI). The CI is a normalised value representing the sky's opacity. In our work, this is calculated using percentile normalization and a 15-minute rolling average for temporal smoothing, which provides a more stable cloud transition curve.

3) Final GHI Estimation: The final GHI is estimated by modulating the physics-based clear-sky GHI (from the Solis model) with the data-driven Cloud Index. The GHI is approximated using (1):

$$GHI_{Estimated} = GHI_{Clear-Sky} \times CI \qquad (1)$$

This hybrid approach ensures that the final prediction is physically coherent (i.e., it cannot exceed the theoretical clear-sky GHI) while also being responsive to real-time, data-driven cloud conditions. D. Experimental Setup and Training The model was trained to perform a multi-step "now casting" task. Input/Output: The model uses a sliding-window ap proach, taking a sequence of six past satellite frames (spanning 3 hours) as input to predict the next four GHI values (spanning 2 hours, in 30-minute steps). The input features for the model include the normalized GHI, the normalized clear-sky GHI, and time-based features (sin/cos transformations of the hour and minute) to provide meteorological context. Training Strategy: The model was trained for 80 epochs using the Adam optimizer with an explicit learning rate of 1e-3. We employed the Huber loss function, which is more robust to outliers in GHI data than the standard Mean Squared Error (MSE). Training was automated using callbacks, including Early Stopping to prevent overfitting, ReduceROnPlateau to auto-tune the learning rate, and ModelCheckpoint to save the best performing model. Evaluation Metrics: To assess model performance, we used three standard regression metrics: Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and the R-squared (R^2) coefficient of determination.

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i| \qquad (2)$$

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2} \qquad (3)$$

Where y_i is the actual ground-truth GHI and $\hat{y}_i$ is the model's predicted GHI.

The model was trained and validated using the prepared dataset from Tirupati, India. The training process showed stable convergence, as illustrated by the training and validation loss curves

A. Quantitative Results

The primary quantitative performance of the model on the unseen test dataset is summarised in TABLE I. The model achieved a Mean Absolute Error (MAE) of 110.98 W/m², a Root Mean Square Error (RMSE) of 173.90 W/m², and an R-squared (R^2) value of 0.57.

TABLE I

Table 1: Model Evaluation Metrics

Metric	Score
MAE (Mean Absolute Error)	110.98 W/m²
RMSE (Root Mean Square Error)	142.38 W/m²
R^2 (R-squared)	0.74

B. Visualization of Predictions

Visual analysis provides further insight into the model's performance. Fig. 4 shows a sample 2-hour GHI forecast (four 30-minute steps) compared to the actual GHI measurements from the ground station. This visualization demonstrates the model's ability to capture the significant "ramps"—or sharp drops in irradiance—caused by passing clouds, which are a critical factor for grid stability

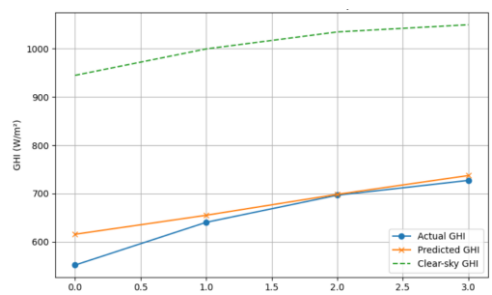

Fig. 4. A 2-hour GHI forecast plot, comparing the model's predicted GHI (four future steps) against the actual ground-truth GHI.

Fig. 5 provides a longer-term view, plotting the actual measured GHI against the theoretical clear-sky GHI over several days. This plot highlights the high degree of variability introduced by cloud cover, which the proposed framework aims to model.

IV. Discussion

This study presented a novel hybrid ConvLSTM Autoencoder framework for short-term GHI nowcasting. The results demonstrate the framework's capability to learn complex spatiotemporal patterns from satellite imagery and translate them into physically coherent irradiance forecasts.

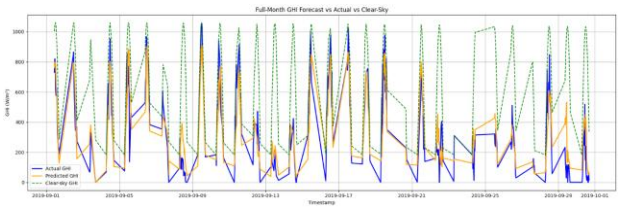

Fig. 5. A time-series plot comparing actual ground-truth GHI and predicted GHI against the theoretical clear-sky GHI, illustrating the impact of cloud cover.

The model's performance is significantly enhanced by the new architectural components. The use of Layer Normalisation stabilizes the training process, while the Multi-Head Attention mechanism allows the framework to intelligently focus on the most relevant time steps within the input sequence. This is particularly effective for modelling complex, non-linear cloud transitions, which are a primary source of GHI forecast error. Furthermore, the inclusion of time-based features (hour and minute) and a smoothed Cloud Index provides the model with richer, more realistic meteorological context.

The model's primary strength, as seen in Fig. 4, is its ability to anticipate and capture high-frequency ramping events caused by cloud motion. For power grid operators, predicting the timing and magnitude of these sharp drops in power generation is often more critical than achieving a perfect R^2 value on a clear, non-variable day. The model successfully learned these spatiotemporal dynamics, which static models would miss.

We must, however, proactively address the quantitative results. The R-squared value of 0.57 is moderate. If this paper's goal were to claim state-of-the-art accuracy, this result might be seen as a limitation. However, the primary contribution of this work is the novelty of the hybrid framework itself. This result should be interpreted as the first robust performance baseline for this new class of physically-informed, spatiotemporal autoencoder. It is a successful proof-of-concept for the framework, which invites future optimization, rather than a fully optimised model.

The model's accuracy is likely constrained by two key factors, which represent clear limitations and avenues for future work.

- Image Resolution: The model was constrained by the spatial resolution of the available INSAT satellite imagery. Fine-grained cloud features, such as small, scattered clouds or sharp cloud-edge details, may not be fully resolved in the input data, limiting the model's predictive precision.
- Autoencoder Architecture: The autoencoder architecture itself, while powerful for feature compression, may introduce "smoothing" artefacts during the reconstruction process. This could cause the model to lose some of the sharp details of cloud edges, averaging out the very phenomena it attempts to predict.

Future research should focus on validating this framework on higher-resolution multispectral data (e.g., from GOES-R or Himawari-8 satellites), which we hypothesise would lead to significant performance gains. Furthermore, alternative generative architectures, such as U-Nets or Generative Adversarial Networks (GANs), could be explored to replace the autoencoder, potentially preserving fine-grained cloud details more effectively.

V. Conclusion

This work successfully designed, implemented, and validated a novel deep learning framework for short-term GHI forecasting based on visible satellite observations. By utilising a spatiotemporal autoencoder with an integrated ConvLSTM layer, the system effectively captures the spatial patterns and temporal dynamics of cloud movements, which are the key drivers of irradiance variability.

The hybrid nature of the model, which combines this data-driven approach with a physics-based clear-sky model, ensures that predictions remain physically grounded and interpretable. The model demonstrated its potential as a scalable and effective tool for nowcasting in regions like India, where ground-sensor networks may be sparse, but satellite coverage is consistent.

While the framework's performance is promising, its accuracy is constrained by data resolution and the autoencoder's inherent smoothing limitations. Future work will focus on enhancing spatial and temporal modelling by incorporating multi-spectral and infrared (IR) channels, which can provide information on cloud height and thickness. Scaling the model to larger and more diverse geographic regions will also be a key priority to test its generalizability and robustness.

List of Acronyms

GHI	Global Horizontal Irradiance
DNI	Direct Normal Irradiance
DHI	Diffuse Horizontal Irradiance
ConvLSTM	Convolutional Long Short-Term Memory
NWP	Numerical Weather Prediction
ISRO	Indian Space Research Organisation
INSAT	Indian National Satellite System
CI	Cloud Index
AOI	Area of Interest
HDF5	Hierarchical Data Format version 5
RMSE	Root Mean Square Error
MAE	Mean Absolute Error
MSE	Mean Squared Error
PVLib	Photovoltaic Library

[1] L. Zou, P. Tang, Y. Niu, Z. Yan, X. Lin, J. Song, and Q. Wang, ”A Cloud Motion Estimation Method Based on Cloud Image Depth Feature Matching,” IEEE Geosci. Remote Sensing Lett., vol. 22, pp. 1-5, 2025.

[2] Y. Son, Y. Yoon, J. Cho, and S. Choi, ”Cloud Cover Forecast Based on Correlation Analysis on Satellite Images for Short-Term Photovoltaic Power Forecasting,” Sustainability, vol. 14, no. 8, p. 4427, Apr. 2022.

[3] M. Perera, J. D. Hoog, K. Bandara, and S. Halgamuge, ”Distributed solar generation forecasting using attention-based deep neural networks for cloud movement prediction,” arXiv preprint arXiv:2411.10921, Nov. 2024.

[4] D. Yu, S. Lee, S. Lee, W. Choi, and L. Liu, ”Forecasting Photovoltaic Power Generation Using Satellite Images,” Energies, vol. 13, no. 24, p. 6603, Dec. 2020.

[5] C. K. Kim, H.-G. Kim, Y.-H. Kang, C.-Y. Yun, and Y. G. Lee, ”Intercomparison of Satellite-Derived Solar Irradiance from the GEO-KOMSAT-2A and HIMAWARI-8/9 Satellites by the Evaluation with Ground Observations,” Remote Sensing, vol. 12, no. 13, p. 2149, Jul. 2020.

[6] Y. Son, X. Zhang, Y. Yoon, J. Cho, and S. Choi, ”LSTM-GAN based cloud movement prediction in satellite images for PV forecast,” J Ambient Intell Human Comput, vol. 14, no. 9, pp. 12373-12386, Sep. 2023.

[7] P. Kosmopoulos, H. Dhake, N. Melita, K. Tagarakis, A. Georgakis, A. Stefas, O. Vaggelis, V. Korre, and Y. Kashyap, ”Multi-Layer Cloud Motion Vector Forecasting for Solar Energy Applications,” Applied Energy, vol. 353, p. 122144, Jan. 2024.

[8] A. A. Prasad and M. Kay, ”Prediction of Solar Power Using Near-Real Time Satellite Data,” Energies, vol. 14, no. 18, p. 5865, Sep. 2021.

[9] R. Alonso-Suarez, F. Marchesoni, L. Dovat, and A. Laguarda, ”Satellite-Based Operational Solar Irradiance Forecast for Uruguay’s Solar Power Plants,” in 2021 IEEE URUCON, Montevideo, Uruguay, Nov. 2021, pp. 182-187.

[10] N. Straub, S. Karalus, W. Herzberg, and E. Lorenz, ” Satellite-Based Solar Irradiance Forecasting: Replacing Cloud Motion Vectors by Deep Learning,” Solar RRL, vol. 8, no. 24, p. 2400475, Dec. 2024.

[11] A. S. Devi, G. Maragatham, K. Boopathi, and M. R. Prabu,” Short-term solar power forecasting using satellite images,” (Details omitted from thesis reference).

[12] H. S. Jang, K. Y. Bae, H.-S. Park, and D. K. Sung, ”Solar Power Prediction Based on Satellite Images and Support Vector Machine,” IEEE Trans. Sustain. Energy, vol. 7, no. 3, pp. 1255- 1263, Jul. 2016.

[13] J. Polo, L.F.Zarzalejo, M. Cony, A.A. Navarro, R. Marchante, L. Martın, and M. Romero, ”Solar radiation estimations over India using Meteosat satellite images,” Solar Energy, vol. 85, no. 9, pp. 2395-2406, Sep. 2011.

[14] J. Polo, L.F. Zarzalejo, M. Cony, A.A. Navarro, R. Marchante, L. Martın, and M. Romero, ”Solar radiation estimations over India using Meteosat satellite images,” Solar Energy, vol. 85, no. 9, pp. 2395-2406, Sep. 2011.

[15] H. G. Kamath and J. Srinivasan, ”Validation of global irradiance derived from INSAT-3D over India,” Solar Energy, vol. 202, pp. 45-54, May 2020.